

Research objective and the next hypothesis

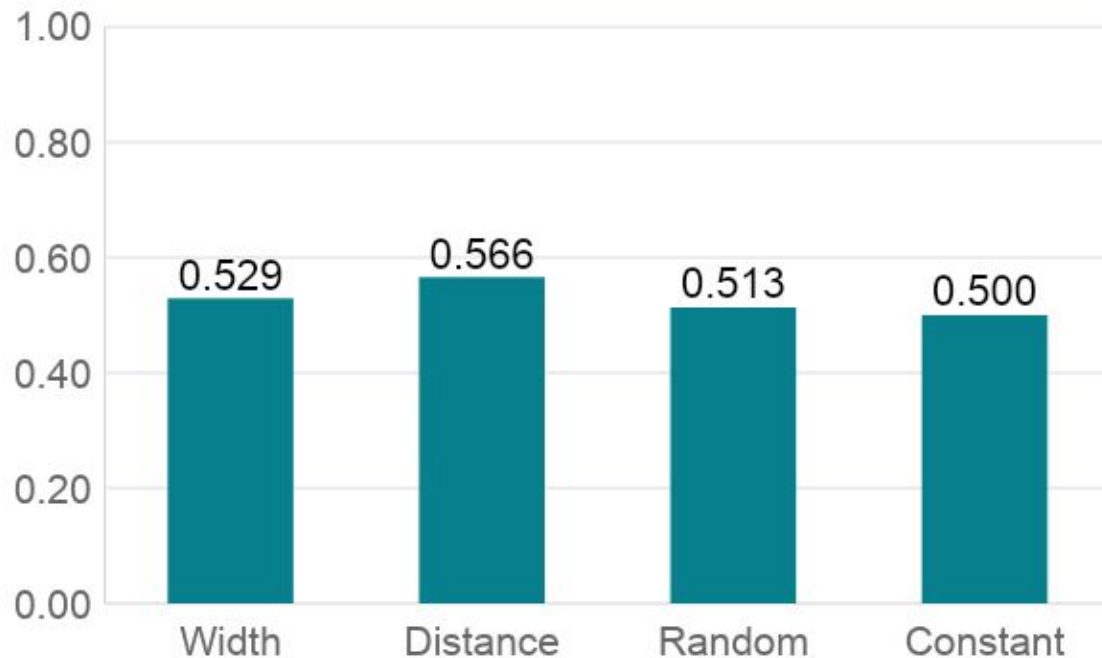
Question	What this study measures	Why it matters
Does BCA operate?	Scale fitting, radius refreshes and consumed multipliers	Separates execution failures from a weak uncertainty signal.
Are bands calibrated?	Fresh response coverage under a declared action shift	Tests whether the stated 90% target still holds.
Do warnings identify harm?	Within-state AUROC and AP against simulator outcomes	Tests usefulness for choosing among actions at one state.
Do choices improve?	Regret among tested actions under a fixed later policy	Connects warnings to local decision quality.
What should IW change?	Weight calibration errors for the target population	Tests a specific mismatch that ordinary CP leaves open.

The study needs more than a return curve

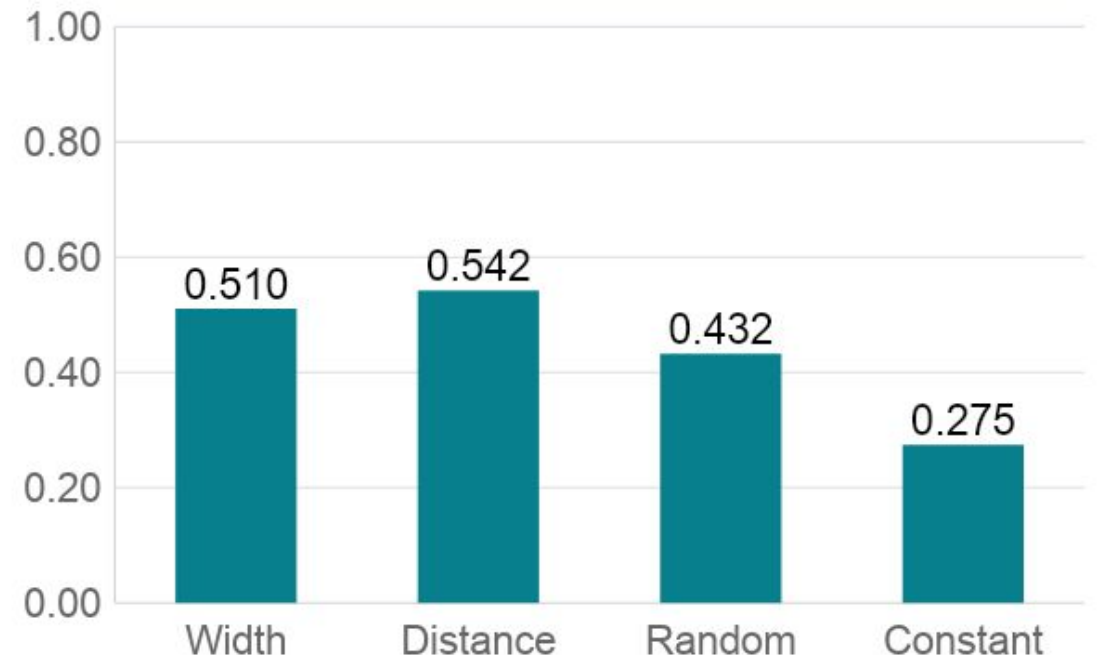
Question	Metric	Evidence available
Are bands reliable?	Fresh empirical residual coverage	Not yet verified
Are warnings useful?	Within-state AUROC and average precision	Completed Hopper pilot
Is unfamiliarity linked to harm?	Support distance versus harm rate	Completed Hopper pilot
Are choices better?	Regret among tested actions	Exploratory pilot readout
Is BCA working during training?	Scale fits, width, radii and penalty dose	Reviewed TD3/ReBRAC; CQL preliminary
What changes in the critic?	Q estimates and Bellman/conservative loss	Training diagnostics; not true-Q error

Warnings: some enrichment, weak separation

AUROC: overall harmful/nonharmful separation



Average precision: quality of top warnings



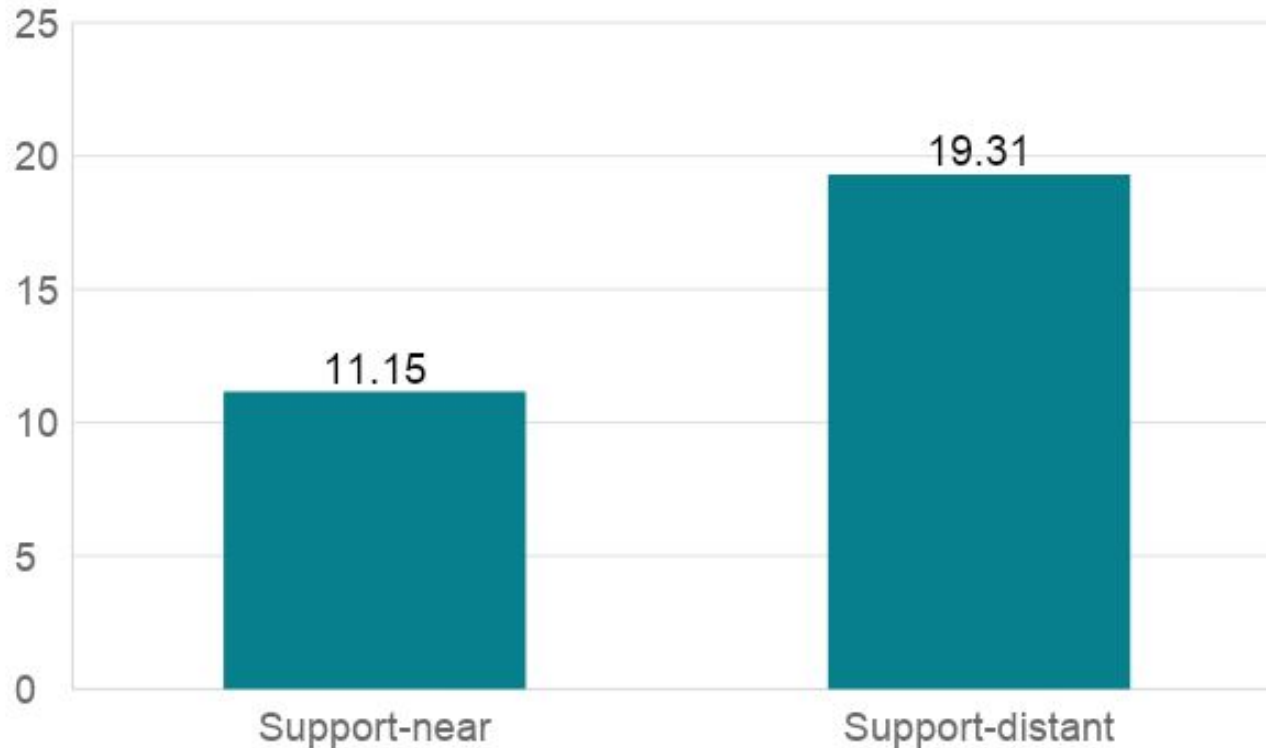
Width carries some signal, but the simple distance warning is stronger here.

Harm ranking has signal, but limited separation

Result and its purpose	Meaning and impact	Improvement to test
AUROC 0.529 Checks harm ordering.	Close to the 0.500 tie baseline. Distance reaches 0.566.	Repeat across seeds and preserved state groups.
AP 0.510 Checks top warnings.	Above constant AP 0.275. Distance is higher at 0.542.	Check whether enrichment persists on new panels.
111/256 eligible panels Shows metric support.	145 panels have one class. Their AUROC/AP remain N/A.	Report valid counts and uncertainty by reset block.
Pooled width AUROC 0.432 Checks across-state mixing.	A different question from choosing actions at one state.	Keep within-state ranking as the primary comparison.
Implication for IW	Useful warnings are incomplete. The cause is still unresolved.	Measure shifted coverage before attributing this to CP.

Unfamiliar actions and harmful actions differ

Observed harmful-action rate (%)



- 231 of 276 harmful actions were still support-near.
- 188 of 233 distant actions were not harmful.
- Novelty is a useful descriptor, not a harm label.

Why test both? BCA must distinguish bad actions, not merely unfamiliar ones.

Distance identifies a harder group, not every bad move

Result and its purpose	Meaning and impact	Improvement to test
Near: 231/2,071 harmful 11.15% of alternatives	Familiar-looking actions can still have poor outcomes.	Keep a near-action control when testing OOD behavior.
Distant: 45/233 harmful 19.31% of alternatives	Higher descriptive harm rate. Group sizes are unequal.	Use declared support bands with comparable sampling.
231/276 harmful are near Checks missed harmful actions.	A distance-only warning would miss much of the observed harm.	Evaluate error bands against outcomes within each band.
188/233 distant are harmless Checks unnecessary caution.	Penalizing all distant actions could discard useful actions.	Track lost good actions alongside detected harm.
Implication for IW	Distance alone does not supply a target/source probability ratio.	Define source and target laws before constructing weights.

Warning quality changes with the situation

Collected states	Eligible / 64	Width AUROC	Distance AUROC
Host · reset	23	0.449	0.501
Host · step 100	34	0.640	0.652
BCA · reset	25	0.557	0.588
BCA · step 100	29	0.471	0.522

Widths vary: 1.35–2.48. No panel has all alternative widths tied.

Changing one global radius cannot repair which action is ranked as riskier.

Warning quality depends on the state group

Result and its purpose	Meaning and impact	Improvement to test
Host reset: 0.449 23 eligible panels	Below the tie baseline in this captured group.	Repeat this group with new seeds and reset blocks.
Host step 100: 0.640 34 eligible panels	The most favorable group in this pilot.	Check whether the advantage survives new data.
BCA reset: 0.557 25 eligible panels	Modest separation here. Same initial reset population.	Preserve shared-reset pairing in uncertainty.
BCA step 100: 0.471 29 eligible panels	Later BCA states do not show the same favorable ordering.	Separate state-visitation shift from action shift.
Widths 1.35–2.48 No all-action ties	Changing width alone does not establish useful ranking.	Test scale quality and query-specific CP separately.

Action choice: regret among tested actions

Mean candidate-set regret (raw reward units; lower is better)



The BCA first action does not reduce average tested-action regret in this pilot.

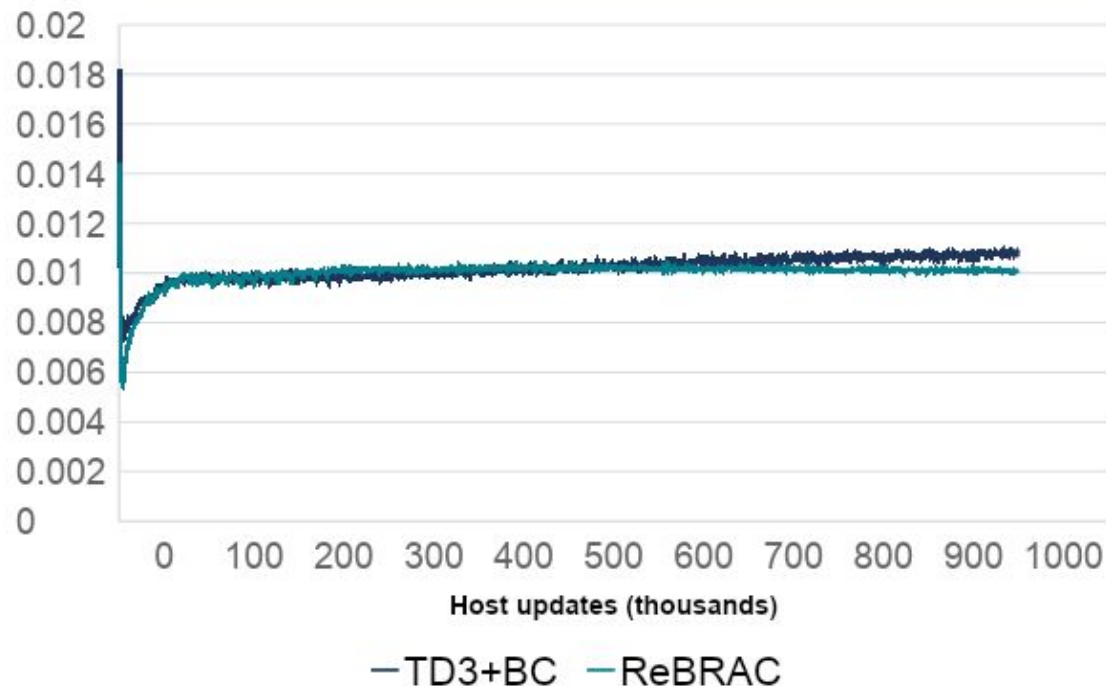
Better average action choice remains unestablished

Result and its purpose	Meaning and impact	Improvement to test
BCA later policy Host 6.333, BCA 6.453	BCA first action has 0.120 more average tested regret.	Repeat paired first-action comparisons across seeds.
Host later policy Host 8.242, BCA 8.490	BCA first action has 0.248 more average tested regret.	Retain both continuations to expose policy dependence.
Why regret matters	Measures missed opportunity within the tested action set.	Report the candidate set and paired uncertainty.
Expectation versus evidence	Expected lower regret with better uncertainty-guided choices.	Check warning quality and how each host uses the band.
Implication for weighted CP	Coverage can improve without improving the selected action.	Require separate evidence for decision benefit.

Calibration is fitting and changing the penalty

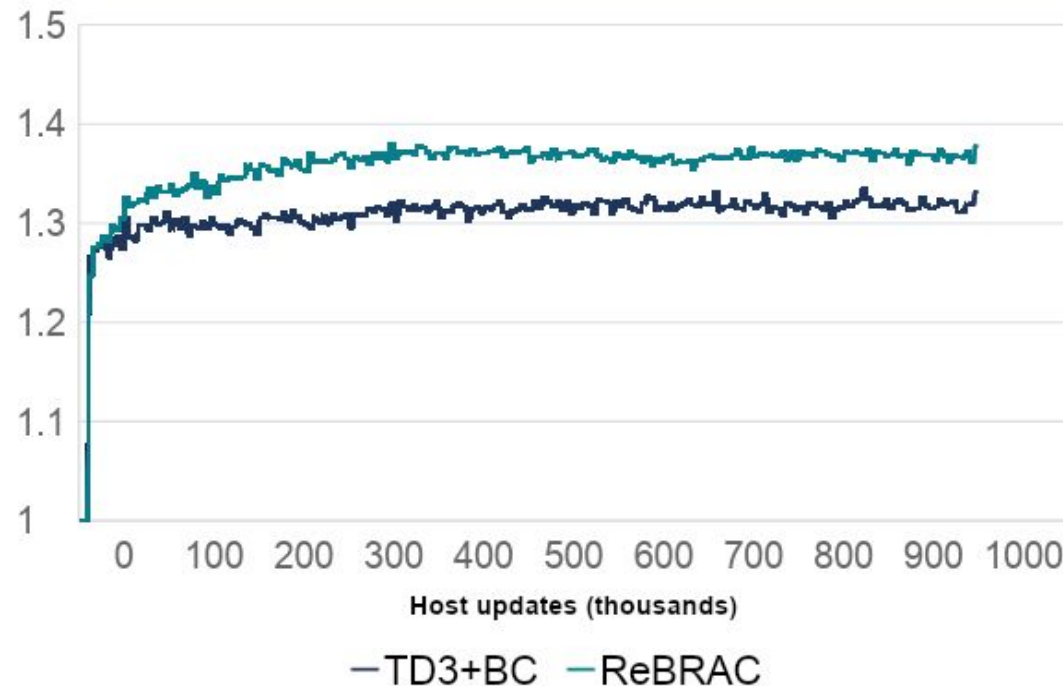
Scale-fitting objective

Block mean objective



Actor behavior-cloning multiplier

Multiplier; 1 = unchanged strength



Both accepted 1,000,000 scale fits. Final block strength: $\times 1.331$ / $\times 1.377$.

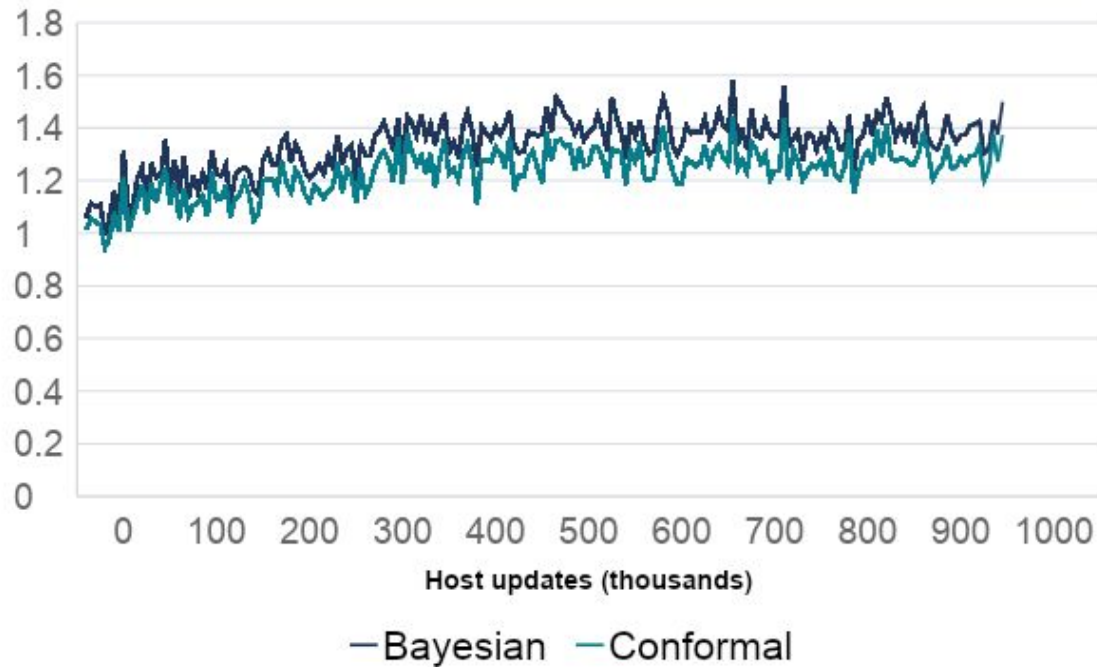
BCA is active, but activity does not prove usefulness

Result and its purpose	Meaning and impact	Improvement to test
1,000,000 accepted fits each Checks calibrator execution.	Both scale learners updated. Inactivity is not the explanation.	Measure fresh residual coverage and warning quality.
Scale objectives 0.01093 / 0.01000	These are fitting diagnostics. Small values do not certify Q.	Test fixed-target errors on an independent bank.
Actor BC dose 1.331 / 1.377	BCA increases imitation strength by about 33% / 38%.	Compare a declared matched fixed-strength control later.
Why dose matters	A useful band still needs an effective host intervention.	Keep host attachment fixed for the first CP comparison.
Implication for IW	Fit weights change the scale. CP weights change the threshold.	Test the weighted threshold before combining both changes.

Bayesian and conformal components are retained

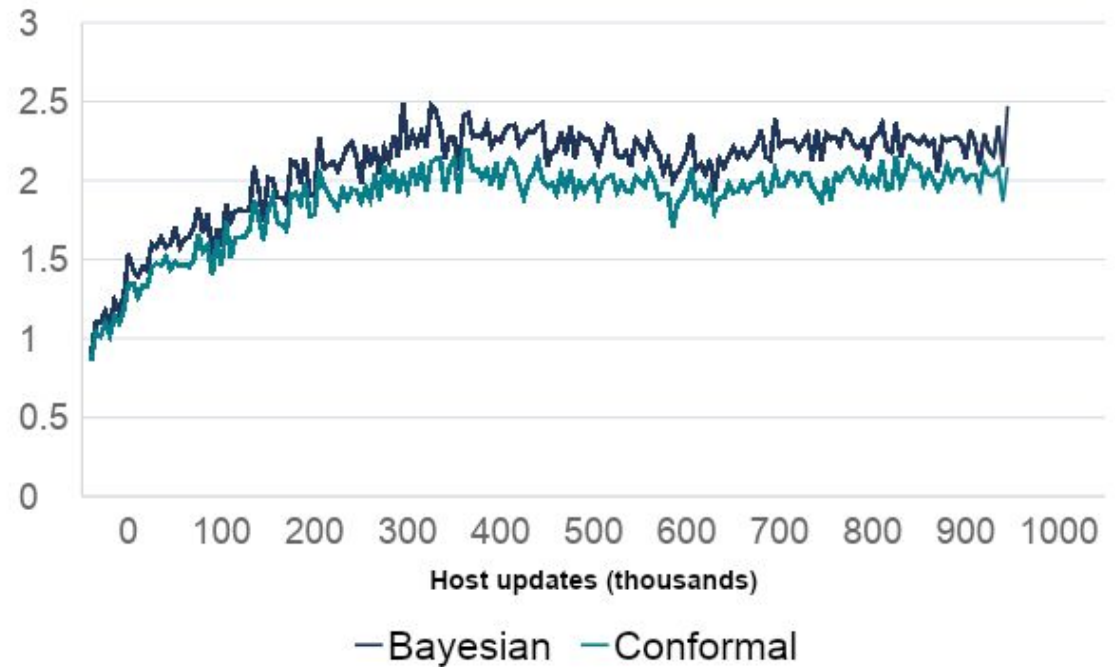
TD3+BC · Walker

Radius in normalized residual units



ReBRAC · Walker

Radius in normalized residual units



Conformal is below Bayesian at 198/198 refreshes in each run.

The floor remains in the algorithm; a larger global radius cannot improve action ordering.

198 refreshes per run. Effective radius is the larger component; this plot is not a coverage test.

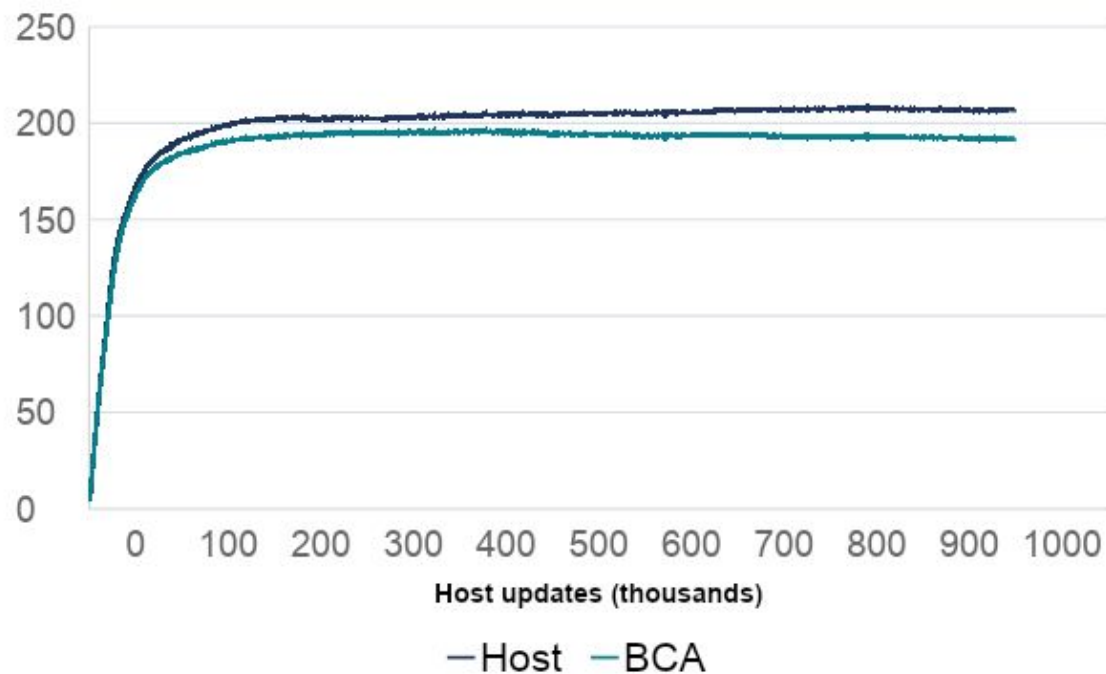
The Bayesian radius controls these observed bands

Result and its purpose	Meaning and impact	Improvement to test
198 refreshes each Checks radius updates.	Both components refresh through the declared schedule.	Verify freshness, snapshots and checkpoint reconstruction.
Conformal below Bayesian 198/198 in both runs	The maximum selects Bayesian at every logged refresh here.	Measure each radius and fresh coverage separately.
Why this matters for IW	A corrected CP radius may still lie below the Bayesian radius.	Check whether the correction changes the consumed band.
Larger global radius	Widens every band without changing action order.	Separate coverage correction from scale/harm alignment.
Expected useful outcome	Adequate coverage with informative finite widths.	Report width and infinity rate alongside coverage.

TD3+BC: inspect Q and Bellman loss

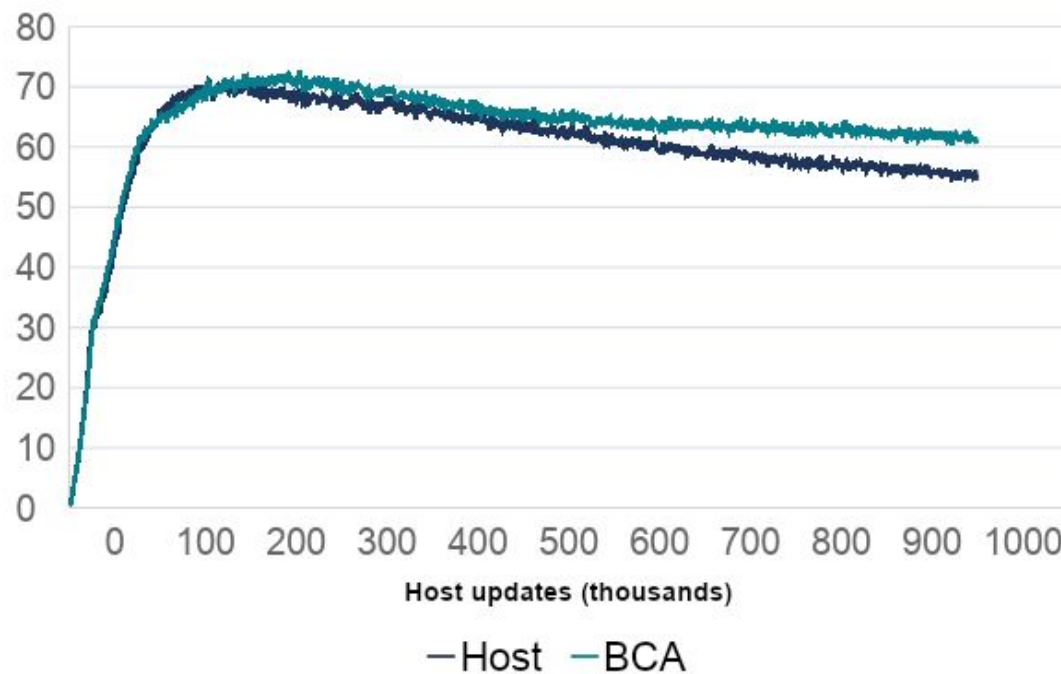
Q1 at proposed actor actions

Block mean Q estimate



Critic fitting loss

Block mean host loss



Final block Q: 207.10 $\rightarrow$ 191.92; critic loss: 55.42 $\rightarrow$ 61.31.

Lower Q does not establish more accurate Q. Larger loss does not identify OOD harm.

One reviewed Walker training seed. Training losses are fitting diagnostics, not labels for unsafe actions.

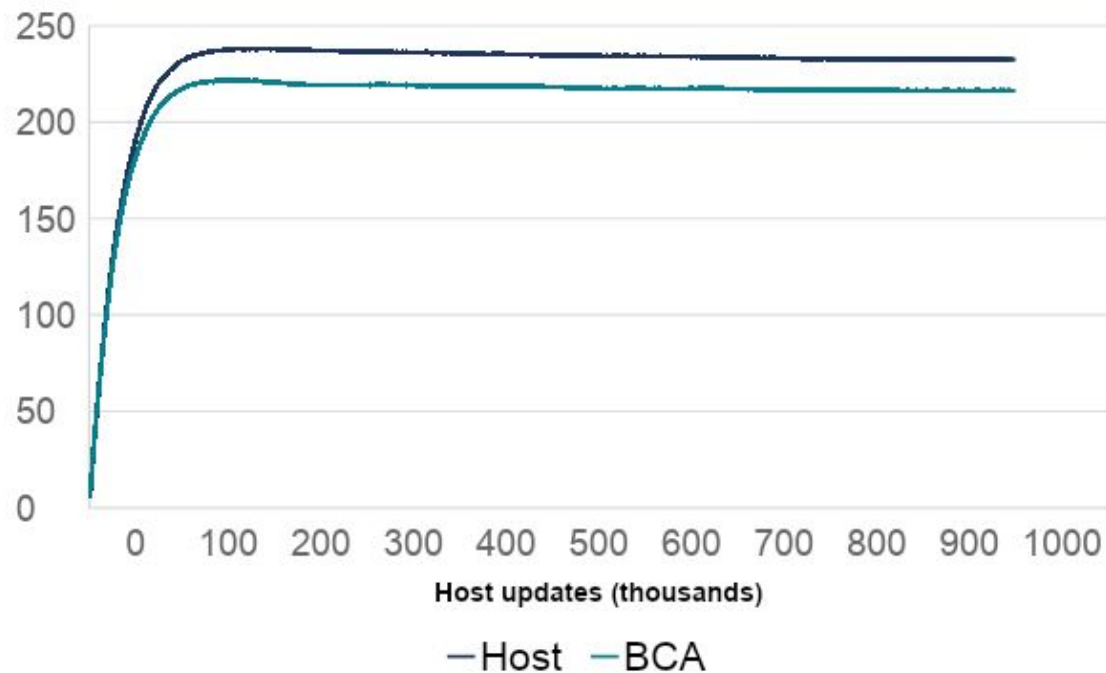
TD3 estimates fall while its fitting loss rises

Result and its purpose	Meaning and impact	Improvement to test
Q1: 207.10 to 191.92 Tracks value predictions.	BCA predicts lower values at proposed actor actions.	Compare against independent value probes at fixed inputs.
Loss: 55.42 to 61.31 Tracks critic fitting.	BCA has higher loss against its current training targets.	Use a shared frozen target and held-out transitions.
Why both matter	Lower Q could reduce optimism or suppress good actions.	Measure prediction error and action harm separately.
Expectation versus evidence	More caution appears in Q. More accurate Q is unverified.	Avoid interpreting lower Q as successful correction.
Implication for IW	Threshold weighting targets response coverage under shift.	It does not make these training targets ground truth.

ReBRAC: inspect Q and Bellman loss

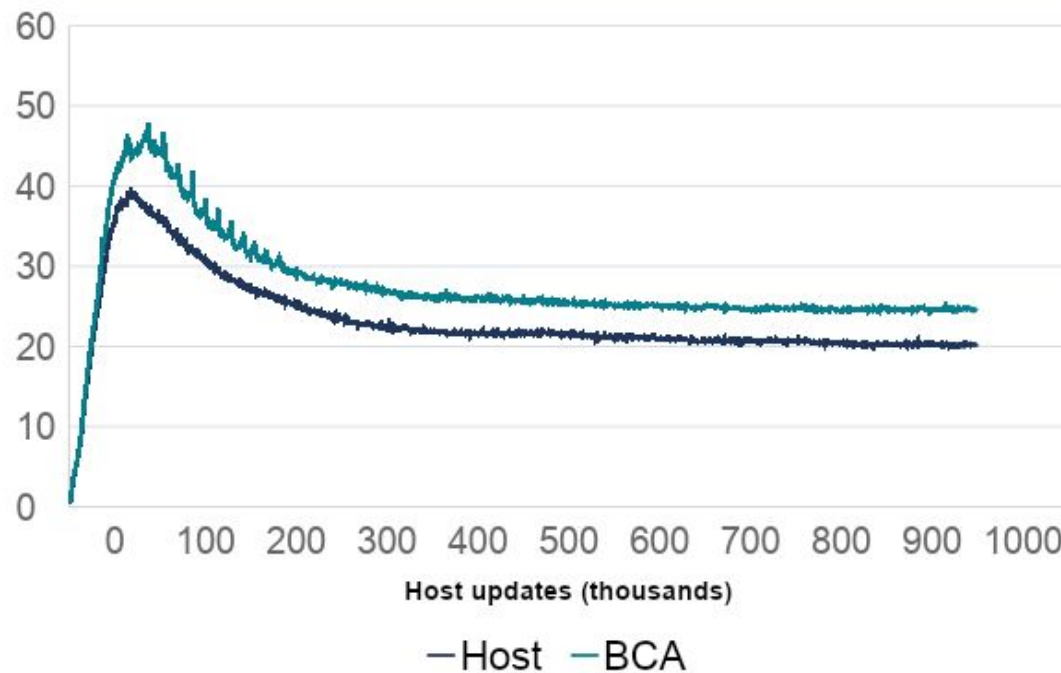
Minimum twin Q at recorded actions

Block mean Q estimate



Critic fitting loss

Block mean host loss



Final block Q: 232.63 → 216.31; critic loss: 20.30 → 24.75.

Lower Q does not establish more accurate Q. Larger loss does not identify OOD harm.

One reviewed Walker training seed. Training losses are fitting diagnostics, not labels for unsafe actions.

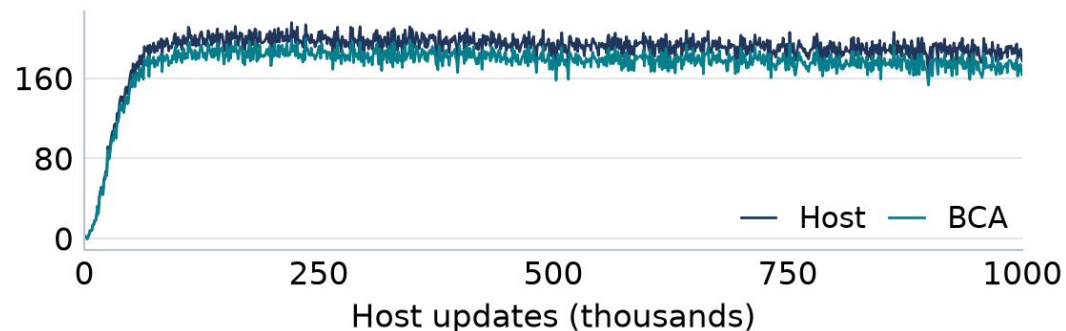
ReBRAC shows the same unresolved value tradeoff

Result and its purpose	Meaning and impact	Improvement to test
Qmin: 232.63 to 216.31 Tracks recorded-action values.	BCA has lower logged Q at dataset actions.	Check shared inputs and independent value evidence.
Loss: 20.30 to 24.75 Tracks critic fitting.	Higher loss accompanies the lower value predictions.	Measure frozen-target error on untouched transitions.
Why the query point matters	Recorded-action Q differs from TD3 proposed-action Q.	Do not rank host optimism using these two log scales.
Expectation versus evidence	The logs show a changed learner. Accuracy remains unresolved.	Connect each diagnostic to the same host's OOD panel.
Implication for IW	Preserve ReBRAC's target and critic BC in the first correction.	Change only the declared calibration calculation.

CQL: inspect its critic and consumed penalty

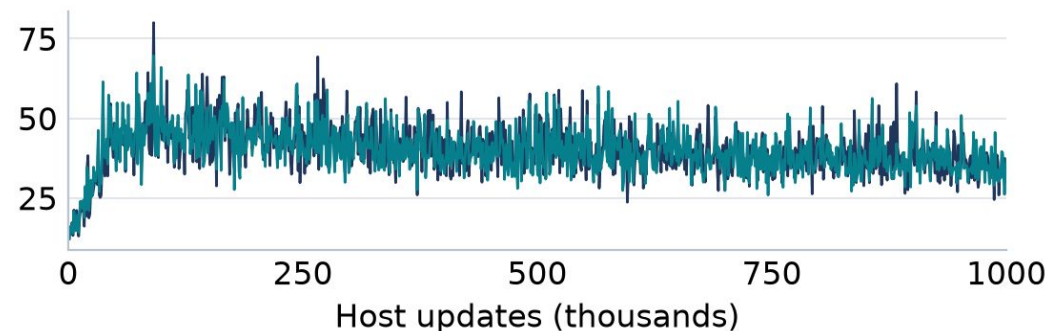
Q1 at recorded actions

Logged Q1



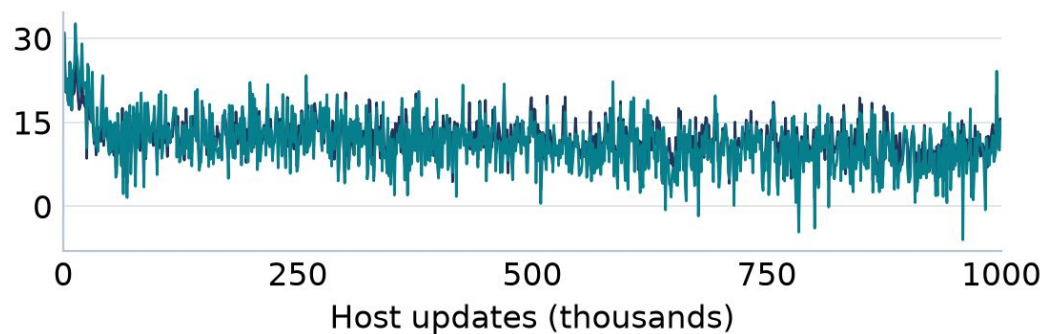
Bellman fitting loss

Logged objective



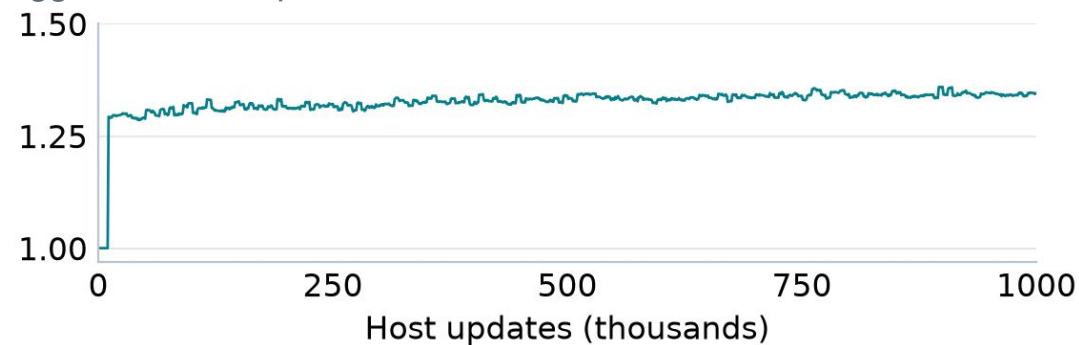
Conservative objective

Logged objective



Conservative-gap multiplier

Logged BCA multiplier



CQL changes its critic, but the mechanism is unverified

Result and its purpose	Meaning and impact	Improvement to test
Q1: 177.15 to 163.12 Tracks recorded-action values.	BCA's logged values are lower. Accuracy remains unknown.	Use fixed inputs and independent value probes.
Bellman loss: 35.60 to 37.20 Tracks target fitting.	The final sampled loss is higher. Targets change during learning.	Compare frozen responses and review checkpoint counters.
CQL objective: 13.68 to 13.21 Tracks conservative training.	The scalar objective changes. It is not a gradient norm.	Measure the consumed loss contribution at fixed inputs.
Gap multiplier: 1.345 Checks BCA's intervention.	One recorded-action band scales the transition's CQL gap.	Audit input alignment before query-dependent integration.
198 refreshes Sparse fitting evidence	Historical numerical radii are absent from this export.	Retain missing values and finish independent review.

Coverage remains an unanswered question

Measure	What it answers	Current evidence
Empirical residual coverage	How often does the band contain its stated target?	Not verified on fresh OOD samples
Band width / sharpness	How much uncertainty does it report?	Pilot widths 1.35–2.48; not coverage
Coverage by support band	Where does calibration fail under shift?	Requires a separate accepted readout
Fit counts and radius refreshes	Did the calibration machinery operate?	Verified for reviewed TD3/ReBRAC runs
True-Q prediction error	Does Q match actual long-run value?	Not supplied by training Bellman loss

Important gap: no verified fresh OOD residual-coverage result. Nominal 90% is a target, not a finding.

What the combined metrics mean for BCA

Evidence	What it establishes	Next step and why
Fits accepted; multiplier changes	BCA is active during training	Evaluate whether that signal is useful
AP enrichment; near-chance AUROC	Some warning information, limited separation	Finish support-balanced, five-seed tests
No lower tested-action regret	No demonstrated choice benefit in the pilot	Check warning-to-host influence separately
Fresh coverage absent	Calibration validity under shift is unresolved	Measure coverage against a declared target
IW only changed fitting previously	Intended weighted-conformal stage was missing	Complete real-data ratios and query-aware host binding

One-seed pilot and reviewed training examples. These are leads to test, not identified causes.

Why these results lead to weighted conformal prediction

What we have learned	Question it raises	Next test and rationale
Scale fits and dose are active.	Are the resulting bands valid for the actions we test?	Measure fresh coverage on a declared target population.
Harm ranking is weak overall.	Does useful calibration degrade as the action population shifts?	Compare coverage and harm within fixed support bands.
Ordinary CP uses source errors.	Do those errors represent target queries adequately?	Weight errors by a justified target/source probability ratio.
Old IW changes scale fitting.	Does threshold weighting fix the missing population correction?	Hold fitting fixed and compare ordinary versus weighted CP.
Regret has not improved.	Would better coverage produce better decisions for this host?	Require separate gains in ranking or regret to claim help.

Current IW fitting leaves the weighted-CP step incomplete

Component	What the current host path does	What weighted CP requires
Scale fitting	Old IW reweights the loss that trains the error scale.	Keep this separate from calibration-threshold weights.
Calibration errors	One group supplies equal masses to the radius refresh.	Target/source mass for each held-out error.
Query correction	The host path supplies query mass 1 and caches one radius.	The actual query ratio and its finite-sample correction.
Weight meaning	Policy density, affinity or advantage are fitting heuristics.	A declared population ratio with justified support.
Independence and output	Calibration feeds later learning. The active cache is scalar.	Fresh frozen-score validation, query-aware cache and infinity.

Weighted CP changes how errors set the band

Step	Calculation	Purpose
Freeze the score	$S = \text{abs}(Y - m(X)) / \text{sigma}(X)$ $X = (\text{state}, \text{action})$	Compare target errors on a common positive scale.
Weight calibration errors	$w(X) = \text{target density} / \text{source density}$	Give more mass to errors more relevant to target queries.
Include the query	Threshold = $0.9 \times (\text{sum of bank weights} + w(\text{query}))$	Account for the unseen query in finite samples.
Find the conformal radius	First sorted error reaching the cumulative mass threshold	If no error reaches it, the radius is infinity.
Retain both components	$R = \max(R_{\text{CP}}(\text{query}), R_{\text{Bayes}})$ Half-width = $\text{sigma}(\text{query}) \times R$	Keep the conformal floor and Bayesian component visible.

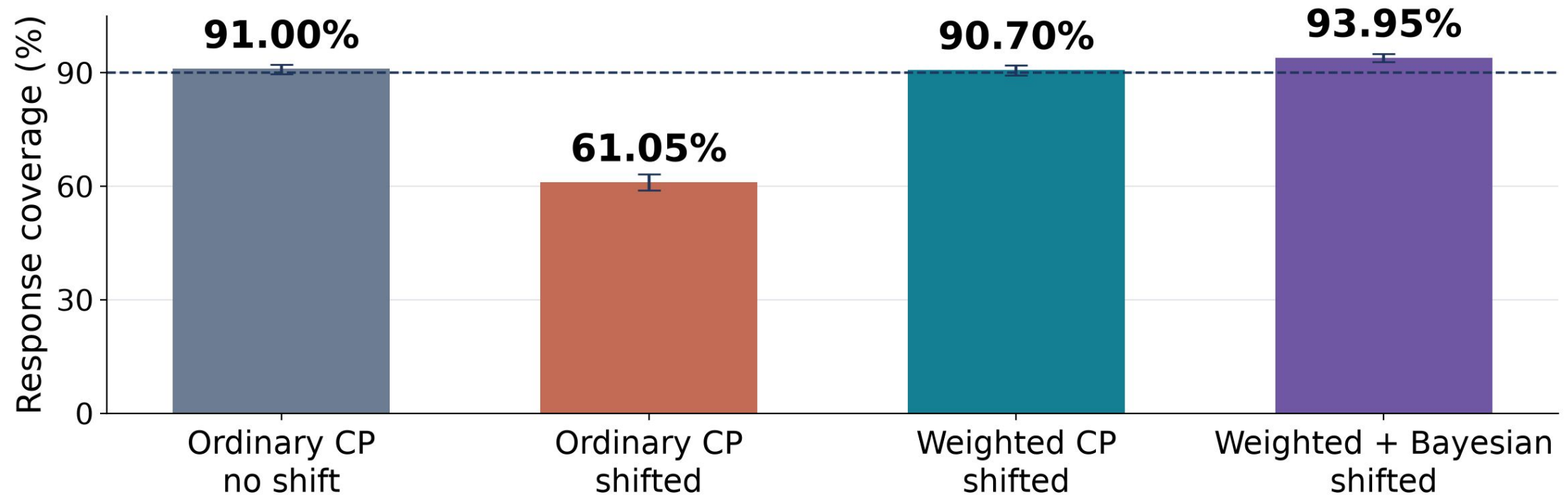
A valid ratio needs a defined population

Requirement	Why it matters here	Implementation decision
Source and target populations	Action shift and changed state visitation need different ratios.	Declare both sampling laws before measuring outcomes.
Support overlap	A point-mass actor lacks a ratio to continuous actions.	Use a justified supported target. Label any neighborhood explicitly.
Same response law given X	Changing target construction changes the quantity covered.	Freeze critic, target, scale and target-action randomness.
Fresh calibration and testing	Training-bank feedback breaks a simple independent-score claim.	Reserve untouched samples and account for trajectories.
Honest extreme-weight handling	Clipping or tempering changes the ratio or target population.	Log weight concentration and preserve infinite bands.

Estimated ratios are an additional approximation. The intended guarantee is marginal response coverage.

Weighted CP restores coverage in the controlled shift

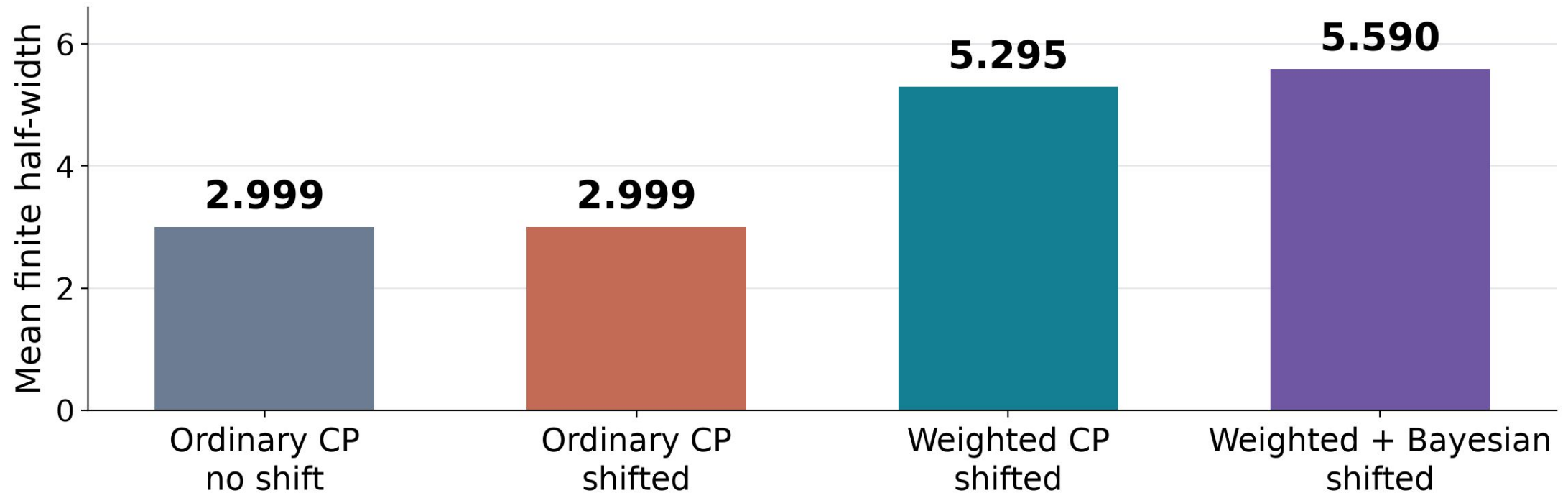
Response coverage (%) with 95% intervals. Dashed line: 90% target.



Shifted coverage rises from 61.05% to 90.70%. Real RL coverage is still unverified.

Coverage recovery comes with wider bands

Mean finite half-width. Scale = 1. Full band length is twice this value.



Wider bands improve coverage here. Their usefulness for decisions still needs testing.

Same 2,000 saved synthetic trials. All bands were finite. Means are descriptive, without uncertainty bars.

A controlled shift validates the correction mechanism

Setting and observed coverage	Why this result matters	Meaning and next step
Ordinary CP on source 91.00% coverage	The source-population control is near the nominal target.	Establishes the starting point for this constructed example.
Ordinary CP after shift 61.05% coverage	Changing population alone can undermine source calibration.	Demonstrates the hypothesized failure in a controlled setting.
Weighted CP after shift 90.70% coverage	Known correct ratios recover approximately nominal coverage.	Validates threshold arithmetic. Real RL ratios remain a gate.
Weighted + Bayesian maximum 93.95% coverage	The retained maximum gives wider bands and extra coverage.	Compare informativeness too. Mean radius 5.590 vs 5.295.
15 reference checks pass 0 infinite bands in these trials	Separate tests require infinity and verify that it stays intact.	Integrate query-aware radii before claiming a host result.

Each host keeps its original BCA attachment initially

Host	Existing place BCA enters	What the integration must preserve
TD3+BC	Detached multiplier on actor behavior cloning	Delayed actor schedule and original host target.
ReBRAC	Detached multiplier on actor behavior cloning	Critic BC and regularized target construction.
CQL	Recorded-transition multiplier on the conservative gap	Original dual gap and actual query point for the band.
IQL	Shrinks capped AWR-weight excess above one	Shared Q/V, fixed gain 1 and original sampling semantics.
All four	Replace scalar-only radius with a query-aware calculation	Versioned checkpoints, finite/infinite behavior and parity.

The first correction isolates calibration. A different host penalty is a separate experiment.

Implementation sequence and comparison plan

Stage	Concrete work	Acceptance evidence
1. Population contract	Freeze score and response. Declare source/target sampling.	Ratios, overlap and sampling assumptions are explicit.
2. Query-aware implementation	Weighted score bank, query mass, Bayesian path and infinity.	Reference agreement and checkpoint round-trip checks.
3. Frozen-model calibration test	Use untouched calibration/test data with a controlled shift.	Coverage, width, infinity rate and weight concentration.
4. Isolated learning comparison	Standard BCA versus corrected CP. Hold fitting and host loss fixed.	Declared seeds, matched data, unchanged alpha and budgets.
5. Fitting-IW extension	Add fitting IW as a separate predeclared comparison.	Distinguish fitting benefit from threshold correction.

Proposed follow-up stages. Mathematical reference is complete. Four-host weighted-CP integration is not.

What the next results would mean for BCA

Possible result	Scientific conclusion	Next action
Coverage improves and ranking/regret improve	Supports a useful correction for this declared shift.	Replicate across seeds, hosts and supported shifts.
Coverage improves but decisions do not	The correction calibrates without a demonstrated policy gain.	Study scale/harm alignment and host consumption separately.
Coverage improves only with very wide or infinite bands	The method is cautious but may be too uninformative.	Inspect overlap and ratio concentration before tuning.
Coverage remains poor under a valid test contract	The intended correction has not solved calibration here.	Audit ratios, dependence, response law and implementation.
Standard BCA already covers adequately under the shift	Coverage mismatch lacks support as the main limitation here.	Investigate decision alignment without forcing an IW narrative.

Success requires answering both calibration validity and decision usefulness. Return remains a secondary context metric.